

جانا يقول عرف ال pure sound وال complex sound سؤال **مقالی** (كل واحد سؤال)

The difference between pure and complex sound is that pure sound has a constant pitch (frequency) while complex sound has more than one pitch (frequency)

وجانا سؤال تعريف ال sound digitization **مقالی**

Digitization is the process of converting information into a digital format.

وجاء سؤال اذكر الثلاث خطوات من ال digitization **مقالی**

1. Sampling

2. Quantizing

3. Coding

اظن ذا جاء بس بسيط ماييغاله، الاجابه تكون 16 عشان $N = 4$ و $2^4 = 16$. الباقي مو مهم

Quantizing

- The relation between the number of quantization levels (M) and the number of bits (N) in every code is given by: $M = 2^N$
- To determine M you need to know the signal range and number of bits in every level.
- Ex. If the signal ranges from +8V to -8V and $N = 4$ then the quantization levels should be assigned as shown

Voltage value	Binary code
7.5 V	1111
6.5 V	1110
5.5 V	1101
4.5 V	1100
3.5 V	1011
2.5 V	1010
1.5 V	1001
0.5 V	1000
-0.5 V	0111
-1.5 V	0110
-2.5 V	0101
-3.5 V	0100
-4.5 V	0011
-5.5 V	0010
-6.5 V	0001
-7.5 V	0000

جاء جدول (كله مقالتي طبعاً) يسأل عن ال bitmapped وال vector images وبرامجها والفرق بينها والفايل فورمات حق كل وحده

Name	Bitmapped Images	Vector Images
Difference	Represented by pixels	Represented by geometric shapes
Programs	Photoshop	Acrobat
Common file formats	PNG	SVG

هذا جاء مقالتي، قانون ال 2^N

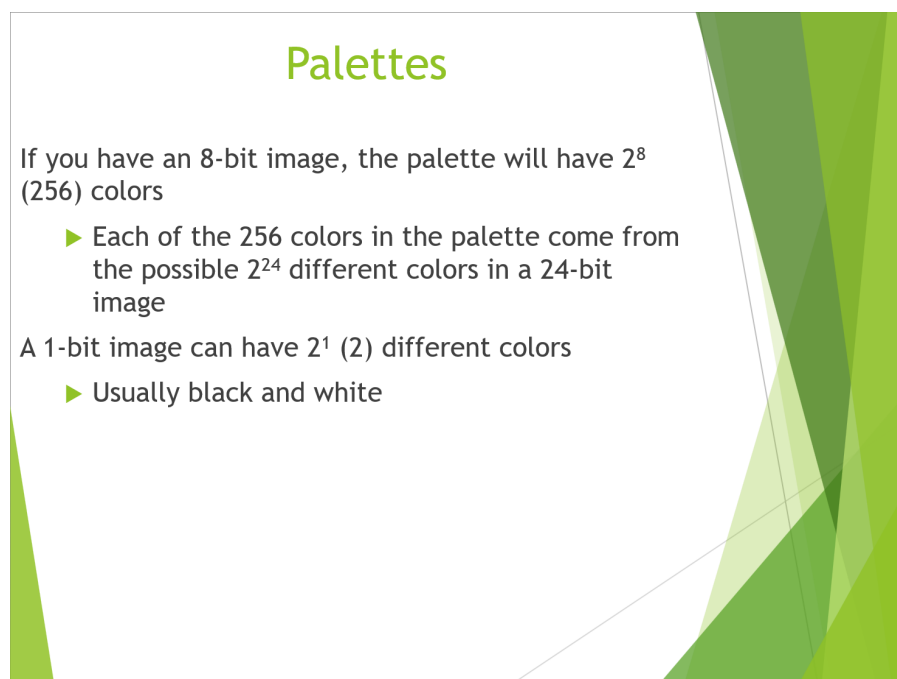
Palettes

If you have an 8-bit image, the palette will have 2^8 (256) colors

- Each of the 256 colors in the palette come from the possible 2^{24} different colors in a 24-bit image

A 1-bit image can have 2^1 (2) different colors

- Usually black and white



هذا جاء كل خطوه حطها سؤال (نفس الأرقام نسخ لصق) **مقالتي**، بس اخر شيء اقسام ال32 على 11.07 ويطلع 2.89 قربته الى صورتين

Example

Calculate the number of megabytes in a color image captured by a digital camera that can produce images with 2272*1704 pixels with bit depth of 24? How many images can be stored in the camera's 32 MB flash memory?

of pixels = 2272 X 1704 = 3,871,488 pixels

of bits = 3,871,488 X 24 = 92,915,712 bits

of Bytes = 92,915,712/8 = 11,614,464 bytes

of MB = 11,614,464/2²⁰=11.07 MB

جاء توصيل

Characteristic	Result
Data integrity	Validity of data is ensured.
Data independence	Input data is kept separate from program data.
Avoiding data redundancy	Data is only entered one time.
Data security	Data is not accessible to unauthorized users.
Data maintenance	Procedures for adding, updating, deleting, and backing up records are in place.

جاء نكتب الكود حق الproject مقالي

COURSES

No	Course-Name	Unit
CIS15	Intro to C	5
CIS17	Intro to Java	5
CIS19	UNIX	4
CIS51	Networking	5
CIS52	TCP/IP Protocols	6



No	Unit
CIS15	5
CIS17	5
CIS19	4
CIS51	5
CIS52	6

select No, Unit
from COURSES

سألنا عن تعريف وخطوات الSDLC مقالي

SDLC provides structure to the development of the goal.

1. Planning
2. Analysis
3. Design
4. Implementation
5. Maintenance

اذكر أنواع الdigital modulation مقالي

1. ASK
2. FSK
3. PSK

جاء error detection and correction مقالتي

Example

Detect and Correct errors in the following stream of bits knowing that the received stream has been encoded using rectangular coding

Received data stream:

11000110 10100010 01101100

Solution

11000110 received correctly

10100010 received incorrectly

To correct the second block XOR the first block with the redundant block

11000110

01101100

10101010 therefore the error was in the fifth bit in the second block

جاء جدول زي الي تحت وقال اذكر كل نوع للتعريف المذكور (الأنواع ماكتبها في الاختبار)
يعتبر السؤال مقالتي

Links computers in a limited geographic area	LAN
Links computer systems a few miles or thousands of miles	WAN
Designed for a city	MAN
Several LANs located in various locations on a college or business campus	CAN
Network of an individual's own personal devices	PAN

اذكر الإيجابيات والسلبيات للnetworking **مقال**

Advantages:

1. Reduced hardware costs
2. Connecting people

Disadvantages:

1. Lack of privacy
2. Security threats

عرف ال LAN topology واذكر أنواعها (اكثر سؤال دفعه 23 ما عرفته) مقال

LAN topologies

- **Network topology**
 - Physical design of a LAN

1. Bus topology
2. Star topology
3. Ring topology

وجات اسئله تعاريف (يعطيك اسم المصطلح فوق وانت تحطه عند التعريف) واسئله صح وخطأ كلها بسيطه ومعلومات بديهية. صح كمان اخر شابت حق ال wired وال wireless ما جاء منه شيء.

بالتوفيق